Node Topology & Driver Models

Comparing Dedicated Docker Nodes (vc-t1, vc-t3) vs. Host Node Inheritance (vc-t2)

1. Docker Driver Model (vc-t1 & vc-t3)

Isolated node containers managed directly by vCluster CLI

Docker Engine Container Space

vcluster.vc-t3 (Control Plane)
Runs API Server, k3s/k8s control components

vcluster.node.vc-t3.t3-worker-1
Mount: ./data/tenant3/n1
Role: Isolated Worker

vcluster.node.vc-t3.t3-worker-2
Mount: ./data/tenant3/n2
Runs: postgres Pod & PVC

t3-worker-3
Mount: ./data/tenant3/n3

t3-worker-4
Mount: ./data/tenant3/n4

Key Characteristic:
- Owns physical Docker node containers.
- Visible in `vcluster list` and `docker ps`.
- Node storage bound via custom `localPathProvisioner.nodePath`.

2. Helm Driver Model (vc-t2)

Nested natively inside host vCluster namespace

my-host vCluster (Host Cluster)

Namespace: tenant2 (on my-host)

Pod: vc-t2-0 (vCluster Control Plane)
Runs as regular Pod inside host vCluster

Pod: redis-78dc58c96d-x-redis-x-vc-t2
Executes directly on my-host's worker-1/2/3 nodes

Pod: coredns-x-kube-system-x-vc-t2
Lazily materializes virtual nodes in vc-t2 when scheduled

Key Characteristic:
- Zero dedicated node containers.
- Invisible in `vcluster list` (requires helm driver context).
- `sync.fromHost.nodes.enabled=true` projects host nodes.